



Premier University, Chittagong

Department of Computer Science & Engineering

Project Proposal

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1. Title

AbilityConnect BD: An AI-Powered Inclusive Disability Assistance Platform for Employment, Healthcare, Accessibility, and Community Support

2. Introduction

People with disabilities in Bangladesh often face numerous challenges in accessing employment opportunities, healthcare services, rehabilitation facilities, volunteer assistance, and government support. Although several organizations provide disability-related services, these services are scattered across different websites and platforms, making them difficult to access. Moreover, many existing systems are not designed with accessibility and personalization in mind. Users frequently struggle to locate relevant information based on their disability type, location, education, or personal needs. To address these challenges, AbilityConnect BD proposes a centralized web-based platform that integrates employment, healthcare, accessibility mapping, volunteer assistance, AI-powered recommendations, learning resources, community engagement, and emergency support into a single accessible system. The platform aims to empower persons with disabilities by connecting them with opportunities and services that enhance their independence, inclusion, and quality of life.

3. Problem Analysis

Persons with disabilities encounter several challenges in their daily lives due to fragmented and inaccessible digital services. Some of the major problems include:

- Lack of a centralized platform for disability-related services.
- Difficulty finding disability-friendly jobs.
- Limited access to rehabilitation centers and healthcare professionals.
- Lack of accessibility information about public places.
- Difficulty requesting volunteer assistance during emergencies.
- Limited awareness of government benefits, scholarships, and NGO services.
- Lack of personalized recommendations based on individual needs.
- Poor communication between users, volunteers, organizations, and healthcare providers.
- Absence of AI-powered assistance for quickly finding relevant resources.

These issues reduce accessibility, increase dependency, and limit opportunities for people with disabilities.

4. Project Overview / Problem Solution

4.1 Existing System Limitations

- Most applications focus on only one service, such as visual assistance, accessibility mapping, healthcare, or employment.
- Users often need to switch between multiple platforms to access different services.
- AI-powered personalized recommendations based on disability type, skills, education, and location are generally unavailable.
- Community-based accessibility reporting and verification features are limited.
- Real-time communication between users, volunteers, organizations, and healthcare providers is often unavailable.
- Most platforms do not integrate NGO services, government benefits, scholarships, healthcare, and volunteer support into a single system.
- Existing systems are not specifically designed to meet the needs of persons with disabilities in Bangladesh.

4.2 Proposed Solution

- Provide a centralized platform for employment, healthcare, education, NGOs, government services, and volunteer support.
- Use Artificial Intelligence to deliver personalized recommendations based on disability type, skills, education, location, and user interests.
- Offer an accessibility map with community reporting and verification.
- Include a volunteer matching system for daily assistance and emergency support.
- Provide real-time chat between users, volunteers, organizations, and healthcare providers.
- Create an accessible, user-friendly, and Bangladesh-focused digital platform that promotes inclusion and equal opportunities for persons with disabilities.

5. Features and Functionalities

5.1 Common Landing Portal & Role-Based Access

All users will first access a common landing page when they visit the AbilityConnect BD platform. This page provides an overview of the platform along with options to Register or Login. During registration, users must select their role to access the appropriate services.

User Roles

- Person with Disability
- Volunteer
- Organization (NGO, Employer, Training Center, or Rehabilitation Center)
- Administrator (Restricted Access)

After successful authentication, the system automatically redirects users to their respective dashboards based on the selected role.

Features

- Common Landing Page
- Role Selection Interface
- Role-Based Registration
- Role-Based Login
- Automatic Dashboard Redirection
- Session Management
- Secure Authentication

5.2 User Management & Authentication

The system provides a secure authentication and user management mechanism for all registered users. It ensures that only authorized users can access role-specific services while protecting user information through secure authentication techniques.

Features

- User Registration
- Login and Logout
- Forgot Password
- Email Verification
- Password Strength Indicator
- Username and Email Availability Checking
- Secure Password Hashing
- Profile Management
- Edit Personal Information
- Profile Picture Upload
- Change Password

5.3 AI Personalized Dashboard

After logging in, each user is redirected to a personalized dashboard based on their role and profile information. The platform uses Artificial Intelligence to analyze user data and provide personalized recommendations that improve accessibility and user experience.

The recommendation engine considers the following information:

- Disability Type

- Skills
- Education
- Location
- User Interests

Based on these factors, the dashboard recommends:

- Accessible Jobs
- NGOs
- Government Services
- Scholarships
- Healthcare Services
- Training Programs
- Volunteer Assistance
- Learning Resources

Person with Disability Dashboard

- Recommended Jobs
- Nearby NGOs
- Volunteer Support
- Doctor Appointments
- Accessibility Reports
- Scholarships
- AI Suggestions

Volunteer Dashboard

- Nearby Help Requests
- Assigned Tasks
- Volunteer History
- Chat Requests
- Community Activities

Organization Dashboard

- Manage Job Posts
- Publish Training Programs
- Manage Scholarships
- Manage Events
- Review Applications

- Respond to User Queries

Admin Dashboard

- Verify Users
- Verify Organizations
- Approve Volunteers
- View Analytics
- Manage Accessibility Reports

5.4 Accessible Job Portal

The Accessible Job Portal enables organizations to publish disability-friendly employment opportunities with detailed accessibility information. It helps persons with disabilities find suitable jobs based on their skills, qualifications, and accessibility requirements.

Job Information

- Job Title
- Company Name
- Salary
- Job Type
- Remote / Onsite
- Accessibility Features
- Flexible Working Hours
- Assistive Technology Support
- Required Skills

Features

- Search Jobs
- Filter Jobs
- Sort Jobs
- Apply Online
- Save Jobs
- Upload Resume
- Track Application Status

5.5 Healthcare & Therapy Finder

The Healthcare & Therapy Finder allows users to discover healthcare professionals and rehabilitation services based on their location and specific needs.

Healthcare Services

- Doctors
- Physiotherapists
- Speech Therapists
- Rehabilitation Centers
- Eye Specialists
- Mental Health Specialists

Features

- Search by Location
- Google Maps Integration
- Appointment Booking
- Ratings & Reviews
- Chat with Doctors

5.6 Accessibility Map

The Accessibility Map is an interactive Google Maps-based module that helps users locate accessible public places and facilities. It also enables the community to report accessibility issues and contribute to improving public accessibility.

Accessibility Information

- Wheelchair Ramp
- Elevator
- Accessible Toilet
- Accessible Parking
- Accessibility Score

Accessibility Reports

Users can report:

- Missing Ramp
- Broken Elevator
- Inaccessible Toilet
- Difficult Entrance

Each report follows the verification process below:

- User Submission
- Community Verification
- Admin Approval

5.7 Volunteer Matching System

The Volunteer Matching System connects persons with disabilities with nearby volunteers who can provide assistance for daily activities and emergency situations.

Assistance Types

- Hospital Visits
- Shopping Assistance
- Travel Assistance
- Document Submission
- Daily Support

Features

- Request Volunteer Support
- Nearby Volunteer Matching
- Accept or Reject Requests
- Task Management
- Real-Time Chat Between User and Volunteer
- Volunteer History

5.8 Organization Services

Organizations can publish various disability-related services to support persons with disabilities through a centralized platform.

Services

- Accessible Jobs
- Training Programs
- Scholarships
- Government Services
- Rehabilitation Programs
- Awareness Events

User Activities

Users can:

- Apply for Jobs and Programs
- Register for Training and Events
- Contact Organizations
- Save Opportunities for Future Access

5.9 Learning Hub

The Learning Hub provides educational resources and skill development opportunities to help persons with disabilities improve their knowledge, digital skills, and career prospects.

Learning Categories

- Braille
- Sign Language
- Computer Skills
- Freelancing
- Digital Literacy

Supported Content

- Videos
- PDF Documents
- Notes
- Quizzes

5.10 AI Recommendation

The AI understand user queries and provide intelligent recommendations based on their needs and profile.

Users can ask questions such as:

"I am visually impaired and looking for a remote job."

The AI can recommend:

- Accessible Jobs
- NGOs

- Scholarships
- Healthcare Services
- Government Benefits
- Training Programs
- Assistive Technologies

5.11 Community Forum

The Community Forum provides a collaborative platform where users can communicate, share experiences, and support one another.

Features

- Ask Questions
- Answer Questions
- Share Experiences
- Like Posts
- Comment on Discussions
- Report Spam or Inappropriate Content
- Community Moderation

5.12 Real-Time Chat System

The Real-Time Chat System enables instant communication between users and service providers, ensuring quick assistance and effective collaboration.

Supported Communication

- Person with Disability ↔ Volunteer
- Person with Disability ↔ Organization
- Person with Disability ↔ Doctor

Features

- Text Messaging
- Image Sharing
- File Sharing
- Real-Time Message Updates
- Online Status Indicator

5.13 Reviews & Ratings

The Reviews and Ratings module allows users to share feedback about services and service providers, helping improve trust, transparency, and service quality.

Review Categories

- Doctors
- Volunteers
- NGOs
- Organizations
- Training Programs

Features

- Star Rating System
- Written Reviews
- Average Rating Calculation

5.14 Admin Dashboard

The Admin Dashboard provides administrators with centralized control over the platform to manage users, monitor activities, and maintain the overall quality and security of the system.

Features

- User Management
- Organization Verification
- Volunteer Approval
- Job and Training Management
- Accessibility Report Verification
- Community Moderation
- Analytics and Reports
- System Settings
- Feedback Management

6. System Design

i. System Architecture

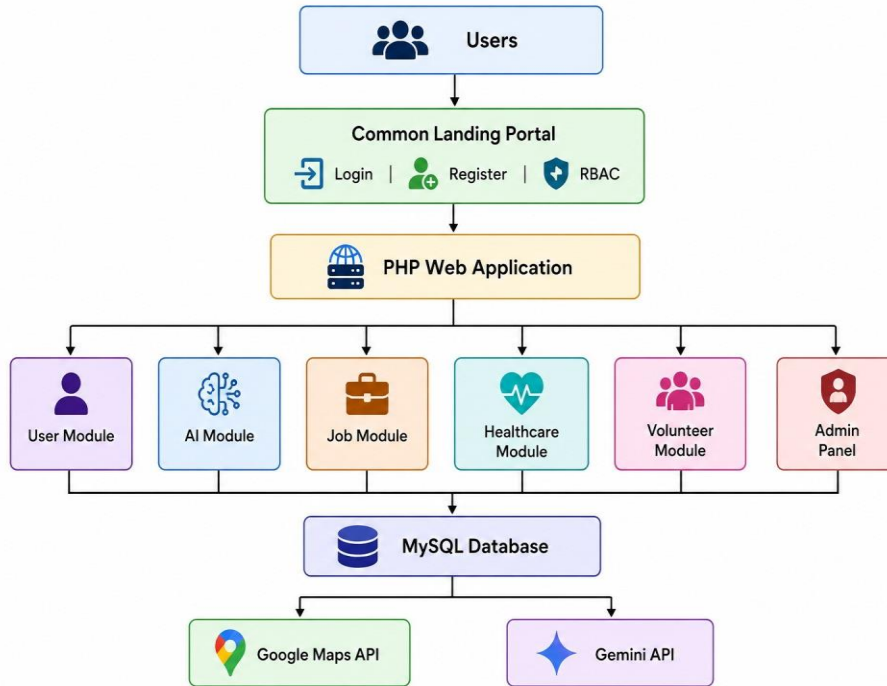


Figure 1: System Architecture of AbilityConnect BD

ii. Database Design (ER Diagram)

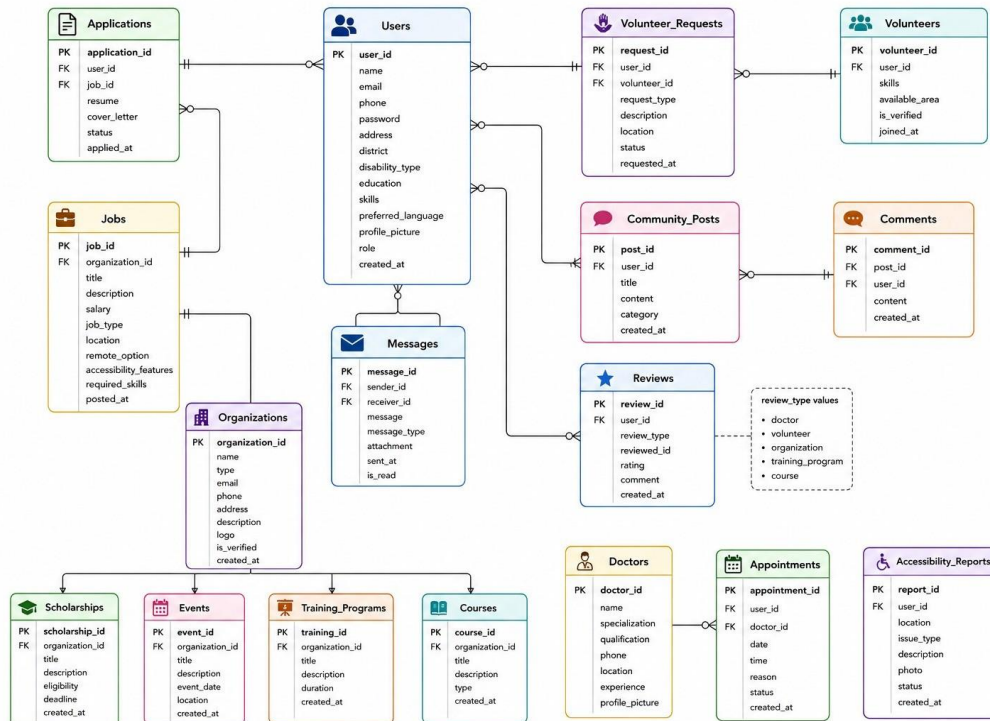


Figure 2: Entity Relationship (ER) Diagram of AbilityConnect BD

7. Technology Stack

7.1 Frontend

- ASP.NET Core MVC (Razor Views)

7.2 Backend

- C#

7.3 Database

- SQL Server

7.4 Tools & Frameworks

- ASP.NET Core MVC
- Entity Framework Core
- Visual Studio , Xampp
- Google Maps API
- Gemini API
- Git & GitHub

8. Implementation Details

The project will be implemented using a modular development approach.

Phase 1

- Authentication
- Role Management
- Database Design

Phase 2

- User Dashboard
- Job Portal
- Organization Module

Phase 3

- Healthcare Module
- Volunteer System

- Accessibility Map

Phase 4

- AI Recommendation System

Phase 5

- Community Forum
- Real-Time Chat
- Testing & Deployment

Security Measures

- Password Hashing
- SQL Injection Prevention
- XSS Protection
- CSRF Protection
- Session Management
- Server-side Validation

9. Group Member Contributions

Member	Contribution
Member 1	Requirement Analysis, Common Landing Portal, User Authentication, Role-Based Access Control, User Profile Management, Database Design, Security Implementation
Member 2	AI Personalized Dashboard, Accessible Job Portal, Organization Services, Learning Hub, Search & Filtering, Job Application Management
Member 3	Healthcare & Therapy Finder, Appointment Booking, Accessibility Map, Google Maps Integration, Accessibility Reporting, Reviews & Ratings
Member 4	Volunteer Matching System, Community Forum, Real-Time Chat, Notification System, Admin Dashboard, Testing, Documentation, GitHub Version Control

10. Conclusion

AbilityConnect BD is an inclusive web platform designed to improve the quality of life of people with disabilities by integrating employment opportunities, healthcare services, accessibility

mapping, volunteer support, AI-powered recommendations, community engagement, and emergency assistance into a single digital ecosystem. The platform promotes accessibility, social inclusion, and independence through modern web technologies and artificial intelligence. It addresses real-world challenges faced by persons with disabilities in Bangladesh and aligns with the objectives of inclusive digital transformation. By combining multiple disability support services within one intelligent platform, AbilityConnect BD has the potential to become a comprehensive solution for enhancing accessibility and equal opportunities.

11. References

- World Health Organization (WHO). *Disability and Health*. <https://www.who.int/health-topics/disability>
- United Nations Convention on the Rights of Persons with Disabilities (UNCRPD). <https://www.un.org/development/desa/disabilities>
- Google Maps Platform Documentation. <https://developers.google.com/maps>
- Google Gemini API Documentation. <https://ai.google.dev>
- PHP Documentation. <https://www.php.net/docs.php>
- MySQL Documentation. <https://dev.mysql.com/doc>
- Bootstrap Documentation. <https://getbootstrap.com/docs>
- Git Documentation. <https://git-scm.com/doc>